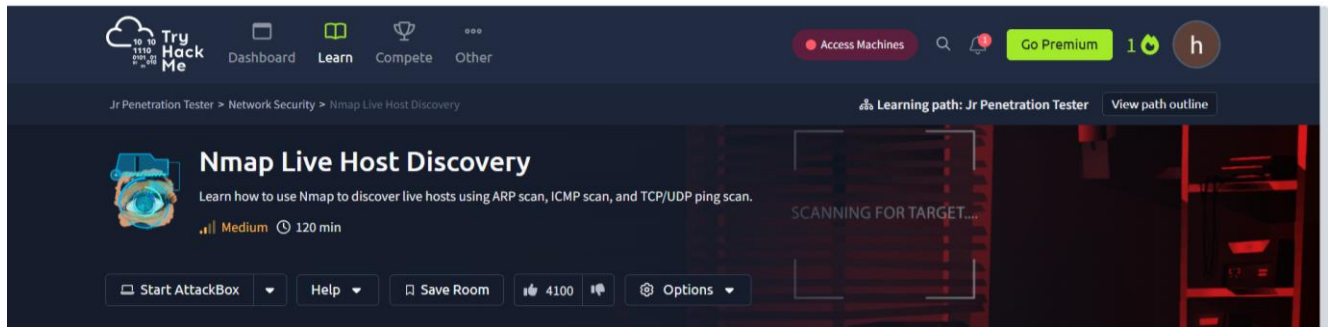


Lab Practical #13:

Prepare a document on any two TryHackMe rooms.

- **Nmap Live Host Discovery :**



1. Introduction :

- Nmap (Network Mapper) is a powerful open-source tool primarily used for network discovery and security auditing. One of its key features is live host discovery, which helps identify active devices on a network. This process is crucial for understanding the layout of a network, detecting unauthorized devices, and performing security assessments.

Task 1  Introduction

When we want to target a network, we want to find an efficient tool to help us handle repetitive tasks and answer the following questions:

1. Which systems are up?
2. What services are running on these systems?

The tool that we will rely on is Nmap. The first question about finding live computers is answered in this room. This room is the first in a series of four rooms dedicated to Nmap. The second question about discovering running services is answered in the next Nmap rooms that focus on port-scanning.

This room is the first of four in this Nmap series. These four rooms are also part of the Network Security module.

1. [Nmap Live Host Discovery](#)
2. [Nmap Basic Port Scans](#)
3. [Nmap Advanced Port Scans](#)
4. [Nmap Post Port Scans](#)

This room explains the steps that Nmap carries out to discover the systems that are online before port-scanning. This stage is crucial because trying to port-scan offline systems will only waste time and create unnecessary noise on the network.

We present the different approaches that Nmap uses to discover live hosts. In particular, we cover:

1. ARP scan: This scan uses ARP requests to discover live hosts
2. ICMP scan: This scan uses ICMP requests to identify live hosts
3. TCP/UDP ping scan: This scan sends packets to TCP ports and UDP ports to determine live hosts.

We also introduce two scanners, `arp-scan` and `masscan`, and explain how they overlap with part of Nmap's host discovery.

Date: 4/03/25

2. Subnetworks :

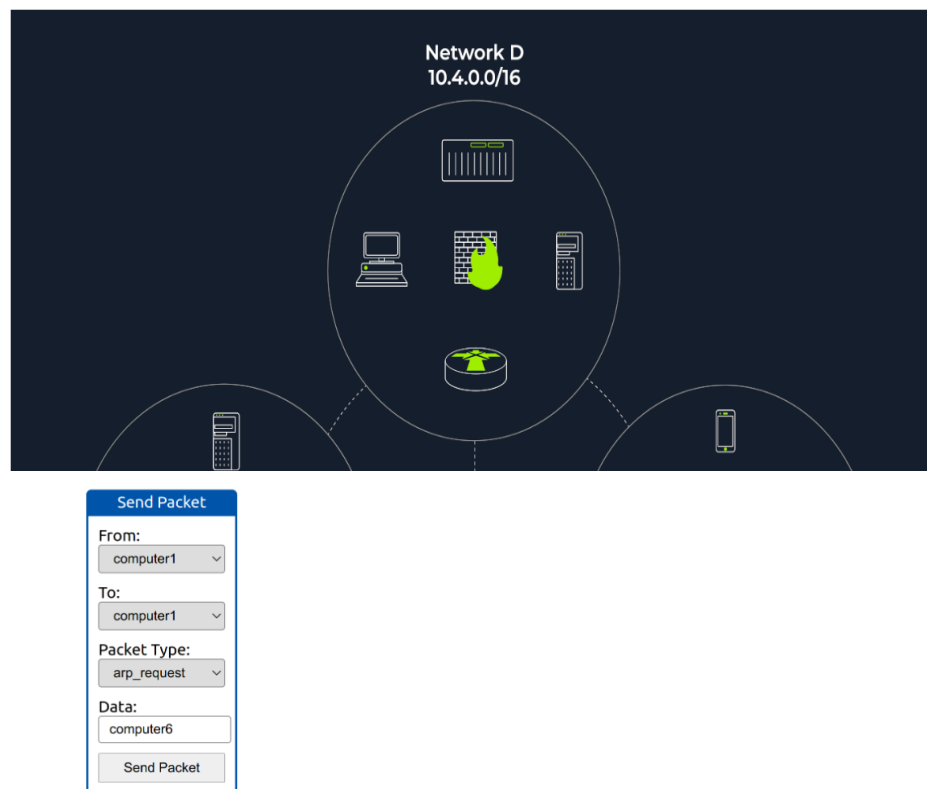
Task 2 Subnetworks



Let's review a couple of terms before we move on to the main tasks. A *network segment* is a group of computers connected using a shared medium. For instance, the medium can be the Ethernet switch or WiFi access point. In an IP network, a *subnetwork* is usually the equivalent of one or more network segments connected together and configured to use the same router. The network segment refers to a physical connection, while a subnetwork refers to a logical connection.

[View Site](#)

In the following network diagram, we have four network segments or subnetworks. Generally speaking, your system would be connected to one of these network segments/subnetworks. A subnetwork, or simply a subnet, has its own IP address range and is connected to a more extensive network via a router. There might be a firewall enforcing security policies depending on each network.



- From computer1
- To computer1 (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: computer6 (because we are asking for computer6 MAC address using ARP Request)

How many devices can see the ARP Request?

4

✓ Correct Answer

🔍 Hint

Did computer6 receive the ARP Request? (Y/N)

N

✓ Correct Answer

3. Enumerating Targets :

Task 3 Enumerating Targets

We mentioned the different *techniques* we can use for scanning in Task 1. Before we explain each in detail and put it into use against a live target, we need to specify the targets we want to scan. Generally speaking, you can provide a list, a range, or a subnet. Examples of target specification are:

- list: `MACHINE_IP scanme.nmap.org example.com` will scan 3 IP addresses.
- range: `10.11.12.15-20` will scan 6 IP addresses: `10.11.12.15`, `10.11.12.16`, ..., and `10.11.12.20`.
- subnet: `MACHINE_IP/30` will scan 4 IP addresses.

You can also provide a file as input for your list of targets, `nmap -iL list_of_hosts.txt`.

If you want to check the list of hosts that Nmap will scan, you can use `nmap -sL TARGETS`. This option will give you a detailed list of the hosts that Nmap will scan without scanning them; however, Nmap will attempt a reverse-DNS resolution on all the targets to obtain their names. Names might reveal various information to the pentester. (If you don't want Nmap to the DNS server, you can add `-n`.)

Launch the AttackBox using the Start AttackBox button, open the terminal when the AttackBox is ready, and use Nmap to answer the following.

Answer the questions below

What is the first IP address Nmap would scan if you provided `10.10.12.13/29` as your target?

10.10.12.8

✓ Correct Answer

🔍 Hint

How many IP addresses will Nmap scan if you provide the following range `10.10.0-255.101-125`?

6400

✓ Correct Answer

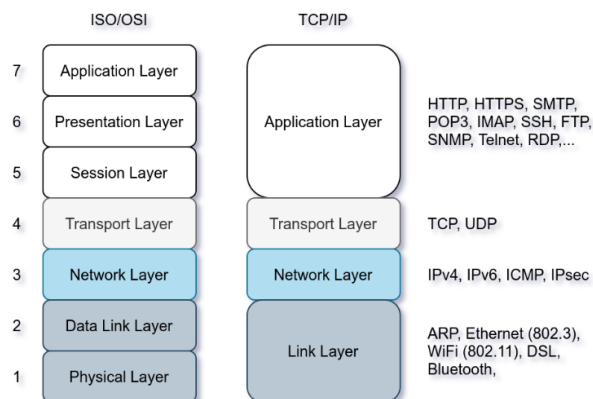
🔍 Hint

4. Discovering Live Hosts :

Task 4 Discovering Live Hosts

Let's revisit the TCP/IP layers shown in the figure next. We will leverage the protocols to discover the live hosts. Starting from bottom to top, we can use:

- ARP from Link Layer
- ICMP from Network Layer
- TCP from Transport Layer
- UDP from Transport Layer





Date: 4/03/25

Send a packet with the following:

- From computer1
- To computer3
- Packet Type: "Ping Request"

What is the type of packet that computer1 sent before the ping?

ARP Request

✓ Correct Answer

What is the type of packet that computer1 received before being able to send the ping?

ARP Response

✓ Correct Answer

How many computers responded to the ping request?

1

✓ Correct Answer

Send a packet with the following:

- From computer2
- To computer5
- Packet Type: "Ping Request"

What is the name of the first device that responded to the first ARP Request?

router

✓ Correct Answer

5. Nmap Host Discovery using ARP :

Task 5 ✓ Nmap Host Discovery Using ARP

How would you know which hosts are up and running? It is essential to avoid wasting our time port-scanning an offline host or an IP address not in use. There are various ways to discover online hosts. When no host discovery options are provided, Nmap follows the following approaches to discover live hosts:

1. When a *privileged* user tries to scan targets on a local network (Ethernet), Nmap uses ARP requests. A privileged user is root or a user who belongs to sudoers and can run sudo.
2. When a *privileged* user tries to scan targets outside the local network, Nmap uses ICMP echo requests, TCP ACK (Acknowledge) to port 80, TCP SYN (Synchronize) to port 443, and ICMP timestamp request.
3. When an *unprivileged* user tries to scan targets outside the local network, Nmap resorts to a TCP 3-way handshake by sending SYN packets to ports 80 and 443.

Nmap, by default, uses a ping scan to find live hosts, then proceeds to scan live hosts only. If you want to use Nmap to discover online hosts without port-scanning the live systems, you can issue nmap -sn TARGETS. Let's dig deeper to gain a solid understanding of the different techniques used.

ARP scan is possible only if you are on the same subnet as the target systems. On an Ethernet (802.3) and WiFi (802.11), you need to know the MAC address of any system before you can communicate with it. The MAC address is necessary for the link-layer header; the header contains the source MAC address and the destination MAC address among other fields. To get the MAC address, the OS sends an ARP query. A host that replies to ARP queries is up. The ARP query only works if the target is on the same subnet as yourself, i.e., on the same Ethernet/WiFi. You should expect to see many ARP queries generated during a Nmap scan of a local network. If you want Nmap only to perform an ARP scan without port-scanning, you can use nmap -PR -sn TARGETS, where -PR indicates that you only want an ARP scan. The following example shows Nmap using ARP for host discovery without any port scanning. We run nmap -PR -sn MACHINE_IP/24 to discover all the live systems on the same subnet as our target machine.

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PR -sn 10.10.210.6/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.00013s latency).
```

Date: 4/03/25

Answer the questions below

We will be sending broadcast ARP Requests packets with the following options:

- From computer1
- To computer1 (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: try all the possible eight devices (other than computer1) in the network: computer2, computer3, computer4, computer5, computer6, switch1, switch2, and router.

How many devices are you able to discover using ARP requests?

3

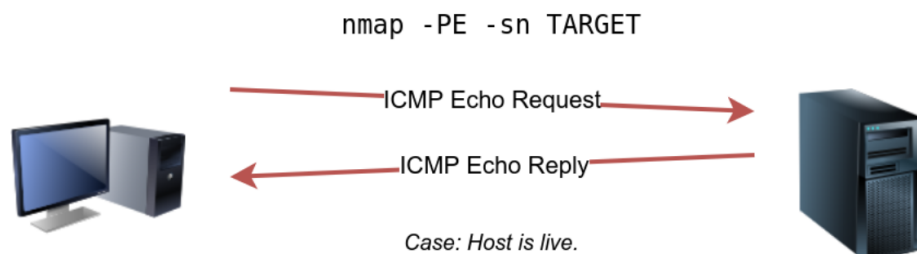
✓ Correct Answer

6. Nmap Host Discovery using ICMP :

Task 6 ✓ Nmap Host Discovery Using ICMP

We can ping every IP address on a target network and see who would respond to our **ping** (ICMP Type 8/Echo) requests with a ping reply (ICMP Type 0). Simple, isn't it? Although this would be the most straightforward approach, it is not always reliable. Many firewalls block ICMP echo; new versions of MS Windows are configured with a host **firewall** that blocks ICMP echo requests by default. Remember that an **ARP** query will precede the ICMP request if your target is on the same subnet.

To use ICMP echo request to discover live hosts, add the option **-PE**. (Remember to add **-sn** if you don't want to follow that with a port scan.) As shown in the following figure, an ICMP echo scan works by sending an ICMP echo request and expects the target to reply with an ICMP echo reply if it is online.



In the example below, we scanned the target's subnet using **nmap -PE -sn MACHINE_IP/24**. This scan will send ICMP echo packets to every IP address on the subnet. Again, we expect live hosts to reply; however, it is wise to remember that many firewalls block ICMP. The output below shows the result of scanning the virtual machine's class C subnet using **sudo nmap -PE -sn MACHINE_IP/24** from the AttackBox.

The scan output shows that eight hosts are up; moreover, it shows their MAC addresses. Generally speaking, we don't expect to learn the MAC addresses of the targets unless they are on the same subnet as our system. The output above indicates that **Nmap** didn't need to send ICMP packets as it confirmed that these hosts are up based on the **ARP** responses it received.

We will repeat the scan above; however, this time, we will scan from a system that belongs to a different subnet. The results are similar but without the MAC addresses.

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PE -sn 10.10.68.220/24

Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 12:16 EEST
Nmap scan report for 10.10.68.50
Host is up (0.12s latency).
Nmap scan report for 10.10.68.52
Host is up (0.12s latency).
Nmap scan report for 10.10.68.77
Host is up (0.11s latency).
Nmap scan report for 10.10.68.110
Host is up (0.11s latency).
Nmap scan report for 10.10.68.140
Host is up (0.11s latency).
Nmap scan report for 10.10.68.142
Host is up (0.11s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap scan report for 10.10.68.222
Host is up (0.11s latency).
Nmap done: 256 IP addresses (8 hosts up) scanned in 8.26 seconds
```

Date: 4/03/25

Answer the questions below

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

-PP

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

-PM

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

-PE

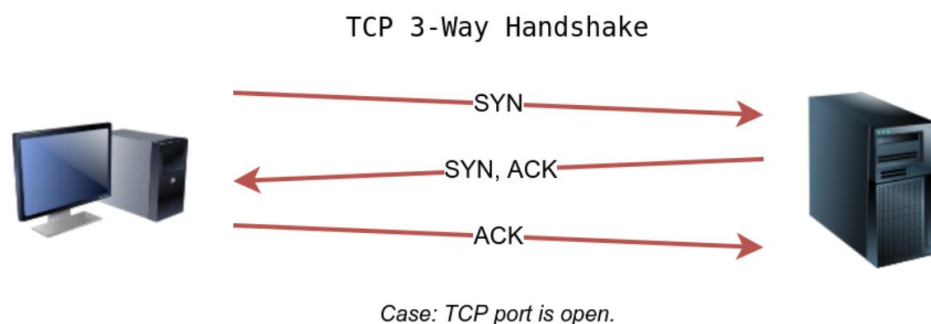
✓ Correct Answer

7. Nmap Host Discovery using TCP and UDP :

• TCP Ping :

TCP SYN Ping

We can send a packet with the SYN (Synchronize) flag set to a TCP port, 80 by default, and wait for a response. An open port should reply with a SYN/ACK (Acknowledge); a closed port would result in an RST (Reset). In this case, we only check whether we will get any response to infer whether the host is up. The specific state of the port is not significant here. The figure below is a reminder of how a TCP 3-way handshake usually works.



If you want Nmap to use TCP SYN ping, you can do so via the option `-PS` followed by the port number, range, list, or a combination of them. For example, `-PS21` will target port 21, while `-PS21-25` will target ports 21, 22, 23, 24, and 25. Finally `-PS80,443,8080` will target the three ports 80, 443, and 8080.

Privileged users (root and sudoers) can send TCP SYN packets and don't need to complete the TCP 3-way handshake even if the port is open, as shown in the figure below. Unprivileged users have no choice but to complete the 3-way handshake if the port is open.

We will run `nmap -PS -sn MACHINE_IP/24` to scan the target VM subnet. As we can see in the output below, we were able to discover five hosts.

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PS -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.16s latency).
Nmap scan report for 10.10.68.125
Host is up (0.089s latency).
Nmap scan report for 10.10.68.134
Host is up (0.13s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 17.38 seconds
```

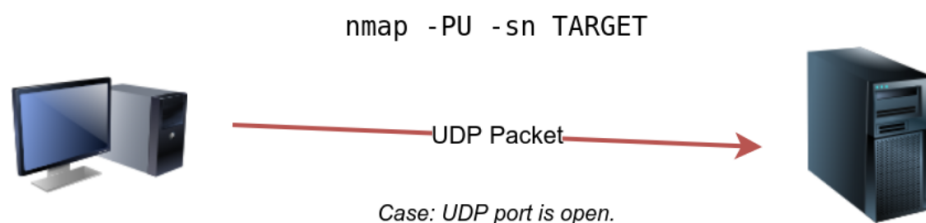
Let's take a closer look at what happened behind the scenes by looking at the network traffic on Wireshark in the figure below. Technically speaking, since we didn't specify any TCP ports to use in the TCP ping scan, Nmap used common ports; in this case, it is TCP port 80. Any service listening on port 80 is expected to reply, indirectly indicating that the host is online.

- UDP Ping :

UDP Ping

Finally, we can use UDP to discover if the host is online. Contrary to TCP SYN ping, sending a UDP packet to an open port is not expected to lead to any reply. However, if we send a UDP packet to a closed UDP port, we expect to get an ICMP port unreachable packet; this indicates that the target system is up and available.

In the following figure, we see a UDP packet sent to an open UDP port and not triggering any response. However, sending a UDP packet to any closed UDP port can trigger a response indirectly indicating that the target is online.



nmap -PU -sn TARGET

The syntax to specify the ports is similar to that of TCP SYN ping and TCP ACK ping; Nmap uses `-PU` for UDP ping. In the following example, we use a UDP scan, and we discover five live hosts.

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PU -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.10s latency).
Nmap scan report for 10.10.68.125
Host is up (0.14s latency).
Nmap scan report for 10.10.68.134
Host is up (0.096s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 9.20 seconds
```

Let's inspect the UDP packets generated. In the following Wireshark screenshot, we notice Nmap sending UDP packets to UDP ports that are most likely closed. The image below shows that Nmap uses an uncommon UDP port to trigger an ICMP destination unreachable (port unreachable) error.

The syntax to specify the ports is similar to that of TCP SYN ping and TCP ACK ping; Nmap uses `-PU` for UDP ping. In the following example, we use a UDP scan, and we discover five live hosts.

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PU -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.10s latency).
Nmap scan report for 10.10.68.125
Host is up (0.14s latency).
Nmap scan report for 10.10.68.134
Host is up (0.096s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 9.20 seconds
```

Let's inspect the UDP packets generated. In the following Wireshark screenshot, we notice Nmap sending UDP packets to UDP ports that are most likely closed. The image below shows that Nmap uses an uncommon UDP port to trigger an ICMP destination unreachable (port unreachable) error.

8. Using Reverse-DNS Lookup :

The syntax to specify the ports is similar to that of TCP SYN ping and TCP ACK ping; Nmap uses `-PU` for UDP ping. In the following example, we use a UDP scan, and we discover five live hosts.

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PU -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.10s latency).
Nmap scan report for 10.10.68.125
Host is up (0.14s latency).
Nmap scan report for 10.10.68.134
Host is up (0.096s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 9.20 seconds
```

Let's inspect the UDP packets generated. In the following Wireshark screenshot, we notice Nmap sending UDP packets to UDP ports that are most likely closed. The image below shows that Nmap uses an uncommon UDP port to trigger an ICMP destination unreachable (port unreachable) error.

9. Summary :

Task 9 Summary

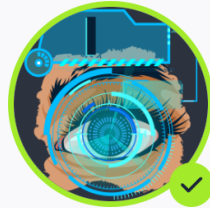
You have learned how ARP, ICMP, TCP, and UDP can detect live hosts by completing this room. Any response from a host is an indication that it is online. Below is a quick summary of the command-line options for Nmap that we have covered.

Scan Type	Example Command
ARP Scan	<code>sudo nmap -PR -sn MACHINE_IP/24</code>
ICMP Echo Scan	<code>sudo nmap -PE -sn MACHINE_IP/24</code>
ICMP Timestamp Scan	<code>sudo nmap -PP -sn MACHINE_IP/24</code>
ICMP Address Mask Scan	<code>sudo nmap -PM -sn MACHINE_IP/24</code>
TCP SYN Ping Scan	<code>sudo nmap -PS22,80,443 -sn MACHINE_IP/30</code>
TCP ACK Ping Scan	<code>sudo nmap -PA22,80,443 -sn MACHINE_IP/30</code>
UDP Ping Scan	<code>sudo nmap -PU53,161,162 -sn MACHINE_IP/30</code>

Remember to add `-sn` if you are only interested in host discovery without port-scanning. Omitting `-sn` will let Nmap default to port-scanning the live hosts.

Option	Purpose
<code>-n</code>	no DNS lookup
<code>-R</code>	reverse-DNS lookup for all hosts
<code>-sn</code>	host discovery only

Date: 4/03/25



Congratulations on completing Nmap Live Host Discovery!!! 🎉

Points earned

🎯 160

Completed tasks

✅ 9

Room type

👤 Walkthrough

Difficulty

📊 Medium

Streak

🔥 1

🗨️ Leave Feedback

Next

- Metasploit : Introduction



Metasploit: Introduction

An introduction to the main components of the Metasploit Framework.

📊 Easy ⌚ 30 min

🔗 Share your achievement

🖥️ Start AttackBox

🆘 Help

🔖 Save Room

👍 6264

⚙️ Options

1. Introduction to Metasploit :

Room progress (41%)

Task 1 Introduction to Metasploit

Set up your virtual environment

To successfully complete this room, you'll need to set up your virtual environment. This involves starting both your AttackBox and Task Machines, ensuring you're equipped with the necessary tools and access to tackle the challenges ahead.

Attacker machine

Status: Connected via Attackbox

Target machine

Status: On [Start Machine](#)

Hey, Congratulations! You submitted the correct answer for task 2 question 4. Keep it up! Metasploit is a powerful tool that can be used for information gathering to post-



2. Main components of Metasploit :

Room progress (41%)

Task 2 Main Components of Metasploit

While using the Metasploit Framework, you will primarily interact with the Metasploit console. You can launch it from the AttackBox terminal using the `msfconsole` command. The console will be your main interface to interact with the different modules of the Metasploit Framework. Modules are small components within the Metasploit framework that are built to perform a specific task, such as exploiting a vulnerability, scanning a target, or performing a brute-force attack.

Before diving into modules, it would be helpful to clarify a few recurring concepts: vulnerability, exploit, and payload.

- **Exploit:** A piece of code that uses a vulnerability present on the target system.
- **Vulnerability:** A design, coding, or logic flaw affecting the target system. The exploitation of a vulnerability can result in disclosing confidential information or allowing the attacker to execute code on the target system.
- **Payload:** An exploit will take advantage of a vulnerability. However, if we want the exploit to have the result we want (gaining access to the target system, read confidential information, etc.), we need to use a payload. Payloads are the code that will run on the target system.

Modules and categories under each one are listed below. These are given for reference purposes, but you will interact with them through the Metasploit console (msfconsole).

Hey, Congratulations! You submitted the correct answer for task 2 question 4. Keep it up! , can be found here.



Date: 4/03/25

Answer the questions below

What is the name of the code taking advantage of a flaw on the target system?

Exploit

✓ Correct Answer

What is the name of the code that runs on the target system to achieve the attacker's goal?

Payload

✓ Correct Answer

What are self-contained payloads called?

Singles

✓ Correct Answer

Is "windows/x64/pingback_reverse_tcp" among singles or staged payload?

Singles

✓ Correct Answer

```
File Edit View Search Terminal Help
root@ip-10-10-216-157:~# msfconsole
This copy of metasploit-framework is more than two weeks old.
Consider running 'msfupdate' to update to the latest version.
Metasploit tip: The use command supports fuzzy searching to try and
select the intended module, e.g. use kerberos/get_ticket or use
kerberos forge silver ticket

# cowsay++
< metasploit >
-----
      \  (oo)\_____/
         (__)        )\
          ||--|| *

      =[ metasploit v6.4.38-dev-
+ -- --=[ 2460 exploits - 1266 auxiliary - 430 post           ]
+ -- --=[ 1468 payloads - 49 encoders - 11 nops             ]
+ -- --=[ 9 evasion                                           ]

Metasploit Documentation: https://docs.metasploit.com/
msf6 > █
```

```
File Edit View Search Terminal Help
|--|| *

+ -- ==[ metasploit v6.4.38-dev- ]
+ -- ==[ 2460 exploits - 1266 auxiliary - 430 post ]
+ -- ==[ 1468 payloads - 49 encoders - 11 nops ]
+ -- ==[ 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > ls
[*] exec: ls

arp.json CTFBuilder Desktop Downloads Instructions new Pictures Postman Rooms Scripts snap thinclient_drives Tools
msf6 > ping 8.8.8.8
[*] exec: ping 8.8.8.8

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
^C
--- 8.8.8.8 ping statistics ---
18 packets transmitted, 0 received, 100% packet loss, time 17385ms

Interrupt: use the 'exit' command to quit
msf6 > Interrupt: use the 'exit' command to quit
msf6 > Interrupt: use the 'exit' command to quit
msf6 > ping -c 1 8.8.8.8
[*] exec: ping -c 1 8.8.8.8

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
--- 8.8.8.8 ping statistics ---
1 packets transmitted, 0 received, 100% packet loss, time 0ms

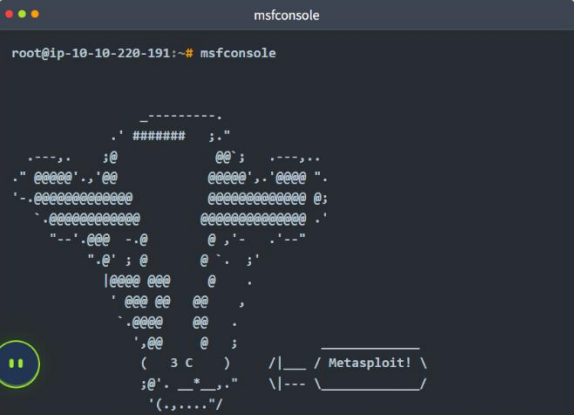
msf6 >
```

3. Msfconsole :

Room progress (58%)

Task 3 ✔ Msfconsole

As previously mentioned, the console will be your main interface to the Metasploit Framework. You can launch it using the **msfconsole** command on your AttackBox terminal or any system the Metasploit Framework is installed on.



```
File Edit View Search Terminal Help
and connected to a database this module will record successful
logins and hosts so you can track your access.

References:
https://nvd.nist.gov/vuln/detail/CVE-1999-0502

View the full module info with the info -d command.

msf6 > info auxiliary/scanner/ssh/ssh_login

Name: SSH Login Check Scanner
Module: auxiliary/scanner/ssh/ssh_login
License: Metasploit Framework License (BSD)
Rank: Normal

Provided by:
todb <todb@metasploit.com>

Check supported:
No

Basic options:
Name Current Setting Required Description
-----
ANONYMOUS_LOGIN false yes Attempt to login with a blank username
BLANK_PASSWORDS false no Try blank passwords for all users
BRUTEFORCE_SPEED 5 yes How fast to bruteforce, from 0 to 5
CreateSession true no Create a new session for every successful login
DB_ALL_CREDS false no Try each user/password couple stored in the current database
DB_ALL_PASS false no Add all passwords in the current database to the session
DB_ALL_USERS false no Add all users in the current database to the session
DB_SKIP_EXISTING none no Skip existing credentials stored in the current database
PASSWORD none no A specific password to authenticate with
```

Date: 4/03/25

Answer the questions below

How would you search for a module related to Apache?

search apache

✓ Correct Answer

Who provided the auxiliary/scanner/ssh/ssh_login module?

toddb

✓ Correct Answer

💡 Hint

4. Working with modules :

Room progress (91%)

Task 4 Working with modules

You can launch the target machine attached to this room to replicate the examples shown below. Any Metasploit version 5 or 6 will have menus and screens similar to those shown here so you can use the AttackBox or any operating system installed on your local computer.

Once you have entered the context of a module using the **use** command followed by the module name, as seen earlier, you will need to set parameters. The most common parameters you will use are listed below. Remember, based on the module you use, additional or different parameters may need to be set. It is good practice to use the **show options** command to list the required parameters.

All parameters are set using the same command syntax:

```
set PARAMETER_NAME VALUE
```

Before we proceed, remember always to check the msfconsole prompt to ensure you are in the right context. When dealing with Metasploit, you may see five different prompts:

- **The regular command prompt:** You can not use Metasploit commands here.

```
st@ip-10-10-XX-XX:~#
```

- **The msfconsole prompt:** msf6 (or msf5 depending on your installed version) is the msfconsole prompt. Any error messages that occur are specific commands to set parameters.

```
File Edit View Search Terminal Help
Name      Current Setting  Required  Description
-----
EXITFUNC  thread           yes       Exit technique (Accepted: '', seh, thread,
LHOST     10.10.216.157    yes       The listen address (an interface may be spe
LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  ---
  0    Automatic Target

View the full module info with the info, or info -d command.
msf6 exploit(windows/smb/ms17_010_eternalblue) > exploit -z
[*] Msf::OptionValidateError One or more options failed to validate: RHOSTS.
msf6 exploit(windows/smb/ms17_010_eternalblue) > back
msf6 > use exploit/windows/smb/ms17_010_eternalblue
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > exploit -z
[*] Msf::OptionValidateError One or more options failed to validate: RHOSTS.
msf6 exploit(windows/smb/ms17_010_eternalblue) > sessions

Active sessions
=====
No active sessions.

msf6 exploit(windows/smb/ms17_010_eternalblue) >
```

Answer the questions below

How would you set the LPORT value to 6666?

set LPORT 6666

✓ Correct Answer

How would you set the global value for RHOSTS to 10.10.19.23?

setg RHOSTS 10.10.19.23

✓ Correct Answer

What command would you use to clear a set payload?

unset PAYLOAD

✓ Correct Answer

What command do you use to proceed with the exploitation phase?

exploit

✓ Correct Answer

Date: 4/03/25

5. Summary :

Task 5 Summary

As we have seen so far, Metasploit is a powerful tool that facilitates the exploitation process. The exploitation process comprises three main steps; finding the exploit, customizing the exploit, and exploiting the vulnerable service.

Metasploit provides many modules that you can use for each step of the exploitation process. Through this room, we have seen the basic components of Metasploit and their respective use.

It would be best if you also had used the ms17_010_eternalblue exploit to gain access to the target VM.

In the following rooms, we will cover Metasploit and its components in more detail. Once completed, this module should give you a good understanding of the capabilities of Metasploit.

Answer the questions below

No answer needed.

No answer needed

✓ Correct Answer

✓ Woop woop! Your answer is correct



Congratulations on completing Metasploit: Introduction !!! 🎉

Points earned

80

Completed tasks

5

Room type

Walkthrough

Difficulty

Easy

Streak

1

Leave Feedback

Next